

Release Summary

Product Name: Vision Studio Spatial Layout Engine

Team Name: Vision Studio

Date: Jun 2, 2026

Key User Stories & Acceptance Criteria

1. Upload a Room or Floor Plan

- 1.1. As a user redesigning my room, I want to upload a floor plan or room image so that I can begin designing with my real space.

Acceptance Criteria

- Given the user is on the upload page, when they select a supported room or floor plan image, the system displays a preview of the uploaded file.
- Given that the user uploads a valid file, when the upload is submitted, the system sends the image into the room parsing/design workflow.
- Given that the uploaded image is processed successfully, when the user continues, the system takes them toward the editable layout/editor experience.
- Given that the uploaded file is unsupported or cannot be processed, when the user attempts to continue, the system should provide an error or fallback path instead of silently failing.

2. View and Edit a 2D Room Layout

- 2.1. As a user redesigning my home, I want to view and edit a 2D room layout so I can begin planning furniture placement and room design.

Acceptance Criteria

- Given a room layout exists, when the user opens the editor, the room appears in a 2D editable workspace.
- When the user is in the editor and interacts with room controls, they can adjust room layout elements such as wall points, dimensions, or layout structure, where supported.
- Given that the user changes the room layout, the editor reflects those changes visually when the layout updates.
- Given that the user is using the editor, when they resize or reposition layout elements, the workspace should maintain a usable and understandable view.

3. Use AI-Assisted Room Guidance

- 3.1. As a user, I want a chatbot or AI-assisted design interface so that I can describe my room goals and receive design guidance.

Acceptance Criteria

- Given that the user opens the chatbot/design assistant, when they type a room goal, style preference, or layout request, then the system accepts the prompt.

- Given that the backend AI service is available, when the user submits a prompt, the system returns a relevant room design or layout-related response.
 - Given that the AI response is shown, when the user reviews it, then the response should help guide layout, furniture, or style decisions.
 - Given that the AI service is unavailable or returns an error when the user submits a request, the system should avoid crashing and should communicate the issue or provide a fallback.
4. Save and Load Project Progress
- 4.1. As a user, I want to save my project progress so that I can continue working on my room design later.
- Acceptance Criteria
- Given the user has a room project open, when they save the project, the current room/layout state is stored.
 - Given that a saved project exists, when the user returns to the app, they can load the saved project.
 - Given that furniture or layout changes have been made, when the project is saved, then the saved state should preserve important layout data such as room information and placed furniture.
 - Given that the database or authentication service is unavailable, when the user attempts to save or load, the system should fail gracefully instead of losing the current editor state without warning.
5. Browse Furniture Options
- 5.1. As a user, I want to browse furniture options so that I can choose pieces that match my room.
- Acceptance Criteria
- Given that the user is in the editor, when they open the furniture catalog, the system displays available furniture items.
 - Given furniture items are shown, when the user views a furniture card, the card includes useful information such as item name, category, dimensions, and a preview image or model.
 - Given the user wants a specific type of item, when they browse categories such as sofa, bed, table, chair, desk, or storage, then matching items should be available.
 - Given that the furniture catalog loads successfully, when the user selects an item, the system allows that item to be used in the room layout.
6. Place, Move, Rotate, and Resize Furniture
- 6.1. As a user, I want to drag and place furniture into my room so that I can test layouts before buying or moving furniture.
- Acceptance Criteria
- Given that the user selects a furniture item, when they place it into the editor, the item appears inside the room layout.

- Given that a furniture item is placed, when the user moves it, then the item updates its position visually.
 - Given that a furniture item is placed, when the user rotates or resizes it where supported, then the editor updates the furniture state.
 - Given that the user saves the project, when the project is reloaded, the furniture position, size, and rotation should be preserved.
7. Detect Furniture Fit and Collision Problems
- 7.1. As a user, I want furniture fit and collision checks so that I can avoid costly layout mistakes.
- Acceptance Criteria
- Given a user places furniture outside the room boundary, when the system validates the layout, it warns the user or prevents the invalid placement where supported.
 - Given that two furniture items overlap, when the system detects the collision, it displays a visual warning or user-facing error message.
 - Given that a furniture item is too large for the room, when the user attempts to place it, the system should indicate that the item may not fit.
 - Given that the user adjusts the item so it fits, when the conflict is resolved, then the warning should no longer block or confuse the user.
8. Receive Furniture or Layout Recommendations
- 8.1. As a user, I want recommendations so that decorating feels easier and more guided.
- Acceptance Criteria
- Given a room layout exists, when the user opens the recommendation area, the system can suggest furniture or layout options based on available room information.
 - Given the user has existing room dimensions or placed items, when recommendations are generated, then they should consider fit, room size, and category, where supported.
 - Given recommendations are shown, when the user reviews them, they should be understandable and relevant to the design workflow.
 - Given that the recommendation system cannot fully validate an item, the system should still rely on fit and collision checks before treating the layout as valid.
9. Auto-Generate a Room Layout
- 9.1. As a user, I want auto-generated room layouts so that I can quickly explore design ideas.
- Acceptance Criteria
- Given the user has a room layout, when they request an auto-generated layout, the system creates a proposed furniture arrangement.

- Given that the generated layout appears, when the user reviews it, they can accept, reject, or manually adjust the result where supported.
 - Given the generated layout includes furniture, when it is placed, then the system should attempt to avoid obvious boundary and collision issues.
 - Given that the generated layout is imperfect, when the user edits it manually, the editor should allow the user to refine the result.
10. View the Room in 3D
- 10.1. As a user, I want to view my designed room in 3D so that I can better understand how the layout will feel.
- Acceptance Criteria
- Given a 2D room layout exists, when the user opens the 3D preview, the system displays a 3D version of the room.
 - Given furniture has been placed in 2D, when the room is viewed in 3D, then the furniture appears in the 3D scene as models or placeholder geometry, where supported.
 - Given the user navigates the 3D view, when they rotate, pan, zoom, or move the camera, then the view updates smoothly enough for review.
 - Given that the 3D view is limited by the current prototype, the system should still provide a useful preview without claiming photorealistic accuracy.
11. Export the Layout
- 11.1. As a user, I want to export my layout so that I can use or share my room design outside of Vision Studio.
- Acceptance Criteria
- Given the user has created a layout, when they choose to export, the system generates a layout export file, if supported.
 - Given that the export includes room and furniture data, when it is downloaded or saved, the export should preserve important layout state.
 - Given 3D export is available, when the user exports the 3D scene, the output should include the room and furniture structure supported by the prototype.
 - Given that an export format is not fully supported, the system should not misrepresent the export as production-ready.

Known Problems:

1. AI and Floor Plan Parsing Limitations
 - Floor plan and room image parsing may vary depending on image quality, lighting, angle, floor plan clarity, and room complexity.
 - Complex room shapes, irregular floor plans, or unclear boundaries may not always be interpreted accurately.
 - Some AI-generated suggestions may be useful for design inspiration but still require validation through room constraints, fit rules, and user review.

2. 2D Layout and Editor Limitations

- The editor supports the core layout workflow, but some advanced editing behaviors may still require manual adjustment.
- More complex room geometry, such as unusual polygons or L-shaped spaces, may not always translate perfectly into all views.
- Some editor interactions may require careful user input and may not fully prevent every invalid state.

3. Furniture Catalog Limitations

- The current furniture catalog is a starter catalog, not a complete real-time provider marketplace.
- Furniture dimensions, images, and models may be simplified or manually curated rather than pulled live from multiple furniture providers.
- Real-time pricing, availability, and provider checkout integration are not included in the final release.

4. Fit and Collision Detection Limitations

- The system includes fit and collision checks, but edge cases may still occur with unusual room shapes, custom dimensions, or dense layouts.
- Collision detection is intended to help users catch obvious layout problems, but it should not be treated as a professional architectural measurement tool.
- User review is still required before making real-world furniture purchases.

5. 3D Visualization Limitations

- The 3D preview is an MVP/prototype visualization and is not fully photorealistic.
- Some furniture may appear as placeholder geometry or simplified 3D models instead of exact real-world product models.
- Translating 2D layouts into 3D furniture rendering was one of the project's major challenges, so some alignment or model fidelity issues may remain.
- Lighting, materials, shadows, and visual polish are functional but not final production quality.

6. Save, Load, and Deployment Limitations

- Save/load depends on Supabase configuration and may not work correctly without the required environment variables and database setup.
- Some tests or flows may require real Supabase credentials to run fully.
- Deployment depends on Vercel, Render, Supabase, and environment variable configuration, so missing or incorrect environment values may cause runtime issues.

- The system has been tested as a class project MVP, not as a production-scale application.

7. Testing and Documentation Limitations

- Automated client and server tests are included, but Python service tests are not currently included.
- Browser-based end-to-end testing is limited or not fully included in the release branch.
- Some known limitations are documented here and/or in the Test Report instead of being fully fixed before release.

Product Backlog:

High Priority for Follow-Up Projects

- User Stories
 1. Improve AI floor plan parsing accuracy
 - 1.1. As a user, I want the system to recognize more complex room shapes and clearer measurements so that my generated room layout is more accurate.
 2. Add stronger manual correction tools
 - 2.1. As a user, I want to easily correct walls, corners, scale, and dimensions so that I can fix AI parsing mistakes without restarting.
 3. Expand real furniture provider integration
 - 3.1. As a user, I want access to furniture from multiple providers so that I can compare more realistic items before making purchases.
 4. Improve 2D-to-3D layout consistency
 - 4.1. As a user, I want my 3D preview to match my 2D layout more precisely so that I can trust the visualization.
 5. Improve 3D furniture model quality
 - 5.1. As a user, I want furniture to appear with more realistic models, materials, and proportions so that the preview feels closer to the real room.
 6. Strengthen collision and clearance rules
 - 6.1. As a user, I want the system to check walkways, door clearance, wall spacing, and furniture spacing so that I can make more practical layout decisions.
 7. Add full project history and versioning
 - 7.1. As a user, I want to save multiple versions of my room design so that I can compare different layout ideas.
 8. Improve export reliability
 - 8.1. As a user, I want exported JSON, GLB, or GLTF files to load reliably in external tools so that I can share or continue my design outside Vision Studio.

9. Add a complete user guide and onboarding flow inside the app
 - 9.1. As a new user, I want step-by-step guidance so that I understand how to upload, edit, furnish, preview, and export my room.
 10. Add more complete automated testing
 - 10.1. As a development team, we want broader automated tests for the Python service, full save/load flow, deployment configuration, and browser end-to-end workflows so that future changes are safer.
- Bug Fixes
 1. Fix edge cases where AI room parsing produces inaccurate or incomplete geometry.
 2. Fix any mismatch between 2D furniture placement and 3D furniture placement.
 3. Improve error handling when Supabase, the backend server, or the Python service is unavailable.
 4. Improve export validation so exported files are clearly marked as successful or failed.
 5. Remove any non-submission temporary files from the release branch.
 6. Add missing environment templates for any included services that require local configuration.
 7. Improve README and setup instructions for Windows users.
 8. Add clearer user-facing messages for unsupported file types, failed AI parsing, failed saves, and failed exports.
 - Future Enhancements
 - Photorealistic rendering
 - AR in-home preview
 - VR walkthrough support
 - Real-time furniture pricing and availability
 - Budget-aware furniture recommendations
 - Style-aware room generation
 - Multi-room project support
 - Real-time collaboration
 - Shopping list or parts list generation
 - Interior designer collaboration tools
 - Real estate virtual staging mode
 - Renovation cost estimator
 - Live web-to-Unity or web-to-external-tool sync

Last Modified: Jun 4, 2026